A FINITELY PRESENTED FLAT MODULE WITH A COMPLETE DEVISSAGE OVER A PRIME IS, AFTER LOCALISING, A DIRECT SUM OF POWERS OF SMOOTH ALGEBRAS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Finiteness of presentation along a complete devissage. Same chapter flat.tex, label remark-finite-presentation, invoked at proof line 3249. Justifies that the first module of the devissage is finitely presented over the first algebra, which is what lets the induction be set up on the first step.
- The finitely presented induction step. Same chapter flat.tex, label lemma-induction-step-fp, invoked at proof line 3251 and claimed separately as S14-algebra-induction-step-fp. Gives that after inverting an element outside the given prime the first map of the devissage is universally injective over the base, which is the reduction that makes the induction hypothesis applicable.
- Flatness of the cokernel of a universally injective map over a domain, and projectivity of the smooth pieces. Algebra, label algebra-lemma-ui-flat-domain (proof line 3257); Same chapter flat.tex, label lemma-fibres-irreducible-flat-projective-nonnoetherian (proof line 3268). The first shows that the cokernel of the first map is flat over the base, so that the tail of the devissage applies to it; the second shows that the smooth algebra appearing in the decomposition is projective over the base, which gives the final projectivity statement.

Source reference. Original lemma 058P: sources/stacks-chow/tags.txt, source lines 6787–6787. Navigation pointer for \ref{algebra-lemma-ui-flat-domain} used inside the counted span (line 3257); the argument that consumes it is retained in the span.

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Source reference. Original lemma 05FT: sources/stacks-flat/pinned-original-source.tex, source lines 2247–2247. Navigation pointer for \ref{lemma-fibres-irreducible-flat-projective-nonnoetherian} used inside the counted span (line 3269); the argument that consumes it is retained in the span.

Source reference. Original lemma 05IA: sources/stacks-flat/pinned-original-source.tex, source lines 3141–3141. Navigation pointer for \ref{lemma-induction-step-fp} used inside the counted span (line 3251); the argument that consumes it is retained in the span.

Source reference. Original remark 05HY: sources/stacks-flat/pinned-original-source.tex, source lines 1685–1685. Navigation pointer for \ref{remark-finite-presentation} used inside the counted span (line 3249); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $R \to S$ be a ring map of finite presentation. Let N be a finitely presented S-module flat over R. Let $\mathfrak{r} \subset R$ be a prime ideal. Assume there exists a complete dévissage of $N/S/R$ over $\mathfrak{r}$. Then there exists an $f \in R$, $f \notin \mathfrak{r}$ such that*

$$N_f \cong B_1^{\oplus r_1} \oplus \ldots \oplus B_n^{\oplus r_n}$$

as R-modules where each B_i is a smooth R_f-algebra with geometrically irreducible fibres. Moreover, N_f is projective as an R_f-module.

Proof. Let $(A_i, B_i, M_i, \alpha_i)_{i=1,\ldots,n}$ be the given complete dévissage. We prove the lemma by induction on n. Note that the assertions of the lemma are entirely about the structure of N as an R-module. Hence we may replace N by M_1, and we may think of M_1 as a B_1-module. See Remark 05HY in order to see why M_1 is of finite presentation as a B_1-module. By Lemma 05IA we may, after replacing R by R_f for some $f \in R$, $f \notin \mathfrak{r}$, assume the map $\alpha_1 : B_1^{\oplus r_1} \to M_1$ is R-universally injective. Since M_1 and $B_1^{\oplus r_1}$ are R-flat and finitely presented as B_1-modules we see that $\mathrm{Coker}(\alpha_1)$ is R-flat (Algebra, Lemma 058P) and finitely presented as a B_1-module. Note that $(A_i, B_i, M_i, \alpha_i)_{i=2,\ldots,n}$ is a complete dévissage of $\mathrm{Coker}(\alpha_1)$. Hence the induction hypothesis implies that, after replacing R by R_f for some $f \in R$, $f \notin \mathfrak{r}$, we may assume that $\mathrm{Coker}(\alpha_1)$ has a decomposition as in the lemma and is projective. In particular $M_1 = B_1^{\oplus r_1} \oplus \mathrm{Coker}(\alpha_1)$. This proves the statement regarding the decomposition. The statement on projectivity follows as B_1 is projective as an R-module by Lemma 05FT. □